

OneFin Outcome Platform - Deployment Options Evaluation: On-Premises vs AWS

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1. Executive summary

Recommendation: deploy OneFin Phase 1 on-premises with a cloud-portable design, and revisit AWS at the 12-month review. Both options meet the requirement comfortably; the choice turns on integration, approvals and cost of ownership.

Measure	On-premises	AWS
Launch volume supported (300 events/min)	Yes, large headroom	Yes, large headroom
Growth to 50 events/sec without redesign	Yes	Yes
Indicative 3-year incremental TCO	~\$400K	~\$470K
Year 1 infrastructure run cost	~\$46K	~\$32K
Time to production	~10 weeks	~20 weeks
Weighted evaluation score (out of 5)	4.25	3.45
New approvals needed	None beyond standard	Cloud risk assessment, data classification, exit plan

AWS has lower infrastructure cost, but higher onboarding and application-team operating effort outweigh it at this volume. AWS becomes the better choice if an approved cloud pattern already exists, source systems move to AWS, or volume grows well beyond 25 events per second.

Decision requested: approve on-premises deployment for Phase 1, and approve the validation steps in Section 12 to firm up cost figures before ARB sign-off.

2. Context and scope

OneFin is an outcome state and decisioning platform, not a processing platform. It receives status events from finance systems, tracks each business outcome, and decides READY, BLOCKED or DELAYED. The heavy work (P&L, reconciliation, ledger posting) stays in the source systems.

In scope	Out of scope
Event ingestion from Motif, GMIS, RAM, SAP, Axiom and future sources	Financial computation, reconciliation, data transformation
Outcome state, dependency tracking, rule evaluation	Replacing any source system workflow
SLA timers, escalation, notification, downstream propagation	Bulk data movement or batch processing
Audit trail, replay, React UI with real-time updates	Reporting warehouse (can consume OneFin events later)

Because compute demand is low, the deployment decision rests on reliability, control, integration and cost of ownership, not raw capacity.

3. Workload assumptions

OneFin launches at 300 events per minute (5 per second) and is sized to grow to 50 per second without redesign. Volumes below assume 24x7 flow, which overstates totals; real traffic clusters around end of day and T+1 morning, so peak ratio matters more than average.

Tier	Timing	Avg rate	Peak rate (4x)	Events per day	Events per month
T1 Launch	Year 1	300/min (5/s)	20/s	432,000	13.0M
T2 Growth	Year 2	1,500/min (25/s)	100/s	2.16M	64.8M
T3 Design ceiling	Year 3	3,000/min (50/s)	200/s	4.32M	129.6M

Parameter	Assumption
Average event payload	1 KB (canonical status event)

Parameter	Assumption
Stored footprint per event	~3 KB (raw event, decision log, state, indexes)
Hot data in operational DB	90 days: ~120 GB at T1, ~1.2 TB at T3
Archive retention	7 years, compressed ~5x: ~95 GB per year at T1, ~950 GB per year at T3
UI users	~200 concurrent operations and controller users
Burst handling	Queue absorbs short bursts up to 500/s and drains within 15 minutes

Number of tracked outcomes per business day and fan-out per event are not yet confirmed; both affect database sizing more than event rate does.

4. Non-functional requirements

Both options must meet the same targets; these drive design far more than throughput. Targets are proposed and need sign-off from the business owner and operational resilience.

Requirement	Proposed target	Why it matters
Availability	99.9% business hours, 99.5% overall	Controllers rely on READY/BLOCKED status for sign-off
RPO / RTO	RPO 0 for accepted events, RTO 2 hours	A lost event can hide a blocked outcome
Event durability	Every accepted event persisted before acknowledgement	Provable completeness for audit
Idempotency	Duplicate events produce no second state change	Sources retry; networks duplicate
Consistency	Concurrent events on one outcome never corrupt state	Optimistic locking with version column
Latency	P95 under 2 seconds from receipt to decision	Near real-time dashboards and escalation

Requirement	Proposed target	Why it matters
Auditability	Every decision records event, prior state, new state, rule version	Answers "why was this BLOCKED" months later
Replay	Rebuild any business date's outcomes from the event log	Incident investigation, rule change validation
Security	SSO, role-based entitlements, encryption in transit and at rest, privileged access control	Finance data is confidential
Change control	SOX-aligned ITGC: segregation of duties, approved releases, evidence retention	Supports financial reporting controls

5. Target logical architecture (common to both options)

The application is identical on both platforms: Java 21 / Spring Boot 3.5 services, a React UI, and a relational store. Only the infrastructure underneath changes, which keeps the decision reversible.

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flowchart LR
    S[Source systems<br/>Motif GMIS RAM SAP Axiom] --> I[Ingestion API<br/>and consumers]
    I --> B[Durable buffer]
    B --> E[Outcome Decision Engine]
    E --> D[(Outcome store<br/>inbox, state, decision log)]
    E --> O[Outbox]
    O --> A[Escalation, notification,<br/>propagation]
    D --> U[React UI<br/>REST + SSE]
    T[SLA timer service] --> E
    D --> R[Archive<br/>7-year audit]

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The diagram reads left to right: events are buffered, decided, persisted, then acted on.

Platform-neutral patterns required in both options:

- **Inbox table** keyed on event ID for idempotency
- **Optimistic locking** on outcome state (version column)
- **Transactional outbox** so notifications fire only after state commits
- **Append-only decision log** with rule version for audit and replay

- **Buffer abstraction** so Kafka, IBM MQ or SQS can be swapped without touching decision logic

6. Option A: On-premises

Option A deploys OneFin on existing enterprise platforms in the primary and DR data centres, on the same network as all current source systems.

Layer	Component	T1 sizing (per site)
Ingress	Enterprise F5 load balancer (existing)	Shared VIP
Compute	OpenShift / Kubernetes (or RHEL VMs)	Engine 3 pods x 2 vCPU/4 GB; API/UI 2 pods x 1 vCPU/2 GB
Buffer	Enterprise Kafka or IBM MQ (existing)	6 partitions, replication factor 3
Database	Oracle 19c on shared DBaaS, RAC + Data Guard	~150 GB, partitioned by business date
SLA timers	Spring scheduler with DB-backed lock (ShedLock / Quartz)	Runs inside engine pods
Archive	Compressed Oracle partitions or enterprise object storage	~95 GB per year at T1
Observability	Existing Splunk / APM / enterprise monitoring	Existing agents
Security	Enterprise SSO, entitlement system, secrets vault, privileged access	Existing
DR	Warm standby in secondary DC, Data Guard sync	Same footprint, scaled down until failover

Strengths

- Same network zone as Motif, GMIS, RAM, SAP and Axiom: no cross-boundary data flow or new connectivity
- Reuses Oracle 19c skills, existing schemas, controls and support model
- No new third-party risk assessment or cloud data classification approval

- Fastest route to production (Section 11)

Weaknesses

- Capacity changes need tickets and lead time (typically weeks)
- Upgrades, patching and DR tests depend on shared infrastructure team schedules
- Fully loaded chargeback for Oracle RAC and tier-1 storage is higher than managed cloud equivalents
- Less elastic if OneFin grows beyond readiness tracking into higher-volume workloads

7. Option B: AWS

Option B deploys OneFin in the bank's AWS landing zone (us-east-1 assumed), multi-AZ in one region with a pilot-light DR region. Source systems stay on-premises, so every event crosses Direct Connect.

Layer	Component	T1 sizing
Connectivity	Direct Connect + Transit Gateway (enterprise shared)	Existing ports assumed
Ingress	Internal Application Load Balancer	1 ALB, 2 AZs
Compute	ECS on Fargate	Engine 3 tasks x 1 vCPU/2 GB; API/UI 2 tasks x 0.5 vCPU/1 GB
Buffer	SQS FIFO, message group = outcome ID	Batched send/receive
Database	Aurora PostgreSQL, Multi-AZ	2 x db.r6g.large, ~150 GB
SLA timers	EventBridge Scheduler or DB-backed scheduler	Pay per invocation
Archive	S3 (Parquet) with Glacier tiering, Athena for queries	~95 GB per year at T1
Observability	CloudWatch, forwarded to enterprise Splunk / APM	Structured, sampled logs
Security	IAM with SSO federation, KMS customer keys, Secrets Manager, VPC endpoints, GuardDuty, Config	Per account

Layer	Component	T1 sizing
DR	Aurora Global Database secondary region, infrastructure as code for compute	Compute scaled from zero on failover

Strengths

- Managed high availability, patching and backup for database, queue and compute
- Elastic: scaling from T1 to T3 is a configuration change, not a procurement
- Lower infrastructure run cost at list price (Section 8)
- Aligns with long-term cloud strategy if one exists

Weaknesses

- All source systems are on-premises: every event depends on Direct Connect, adding a failure point and latency
- Requires cloud risk assessment, data classification approval and a regulator-ready exit plan before go-live
- Oracle to PostgreSQL changes SQL dialect, tooling and DBA skills (RDS for Oracle is possible but needs BYOL licences)
- More operational ownership moves to the application team (infrastructure as code, cloud controls evidence, cost management)

8. Cost model (indicative, 3 years)

Indicative 3-year incremental TCO is about \$400K on-premises and \$470K on AWS, a gap of roughly 18% that sits inside the estimate's +/-25% accuracy. Infrastructure is the smaller part; people and onboarding decide the outcome.

Basis: AWS figures use us-east-1 on-demand list prices (September 2026) with no enterprise discount. On-premises figures use placeholder internal chargeback rates that Infrastructure Finance must replace. Application build cost is identical for both options and excluded. Growth path assumed: T1 in Year 1, T2 in Year 2, T3 in Year 3.

8.1 AWS monthly run cost by tier

Line item (USD per month)	T1 (5/s)	T2 (25/s)	T3 (50/s)
Compute: ECS Fargate	143	250	340
Database: Aurora PostgreSQL Multi-AZ (instances, storage, I/O, backup)	431	950	1,140
Queue: SQS FIFO	20	60	120
Network: internal ALB, VPC endpoints, Direct Connect egress	150	160	180
Observability: CloudWatch logs, metrics, alarms	100	200	350
Security and backup: KMS, Secrets Manager, GuardDuty, Config, AWS Backup, S3 archive	115	180	260
Production subtotal	960	1,800	2,390
DR region (Aurora Global secondary, pilot light)	320	660	860
Non-production (DEV, SIT, UAT, business-hours scheduling)	700	800	900
AWS Enterprise Support allocation (~10%)	200	330	415
Cloud platform chargeback (landing zone, shared network, tooling) - assumption	500	500	500
Total per month	2,680	4,090	5,065
Total per year	32K	49K	61K

8.2 On-premises monthly run cost by tier

Placeholder chargeback rates: container platform \$35 per vCPU-month, tier-1 storage \$0.60 per GB-month per site, Oracle DBaaS (RAC + Data Guard) \$1,500 to \$3,000 per month depending on size.

Line item (USD per month)	T1 (5/s)	T2 (25/s)	T3 (50/s)
Compute: OpenShift, prod + DR + non-prod (28 / 40 / 44 vCPU)	980	1,400	1,540

Line item (USD per month)	T1 (5/s)	T2 (25/s)	T3 (50/s)
Database: Oracle DBaaS prod + DR, 3 non-prod schemas	1,950	2,650	3,450
Messaging: enterprise Kafka / MQ	300	400	500
Storage: tier-1 SAN, both sites	240	780	1,620
Backup	50	120	250
Monitoring: Splunk / APM ingest	150	250	350
Network: F5, firewall, DC services	150	150	200
Total per month	3,820	5,750	7,910
Total per year	46K	69K	95K

8.3 Three-year incremental TCO

People cost uses a blended \$100K per FTE-year (onshore and offshore mix), counting only effort that differs between options.

Cost element (USD)	On-premises	AWS	Notes
Infrastructure run, 3 years	210K	142K	Sections 8.1 and 8.2
Application team operations, 3 years	150K (0.5 FTE)	225K (0.75 FTE)	On-prem ops largely sits with shared infra teams inside chargeback; AWS shifts IaC, cloud controls evidence and cost management to the app team
One-time onboarding	40K	105K	On-prem: environments, DR setup, NFR and pen testing. AWS: cloud risk assessment, data classification, IaC, connectivity, PostgreSQL adaptation, pen test, exit plan
3-year incremental TCO	400K	472K	AWS ~18% higher

8.4 Sensitivity: what changes the answer

Scenario	On-premises 3-yr	AWS 3-yr	Effect
Base case	400K	472K	On-prem ahead by 72K
Mature cloud platform team: AWS app ops 0.5 FTE, onboarding 60K	400K	352K	AWS ahead by 48K
Spare on-prem capacity, marginal cash view (no new infra spend beyond storage)	~220K	472K	On-prem clearly ahead
1-year Savings Plans / reserved instances on AWS (~25% off compute and DB)	400K	~450K	Gap narrows
Dedicated Oracle EE licences needed (no ULA): 2 processors + Partitioning, prod and standby, at 50% discount to list	~570K	472K	AWS ahead unless PostgreSQL is also acceptable on-prem
No AWS landing zone exists: bank-grade platform build of 250K to 500K	400K	720K to 970K	AWS not viable for a single application

Oracle list reference: Enterprise Edition \$47,500 and Partitioning \$11,500 per processor, 22% annual support; a Data Guard physical standby needs the same licences as production.

9. Weighted evaluation scorecard

On-premises scores 4.25 and AWS 3.45 out of 5. The gap comes from time-to-value, integration and regulatory readiness, not from cost or technology. Scores: 1 = poor, 5 = strong.

Criterion	Weight	On-prem	AWS	Rationale
Reliability and event durability	20%	4	4	Both meet targets; AWS has stronger managed HA but adds a Direct Connect dependency for every event
Security and regulatory readiness	20%	5	3	On-prem controls already approved; AWS needs cloud risk assessment, data classification and exit plan
Operational effort	15%	3	4	AWS removes patching and capacity work; on-prem relies on shared team schedules
3-year TCO	15%	4	3	Base case favours on-prem by ~18%, within estimate accuracy
Time-to-value	15%	5	3	~10 weeks vs ~20 weeks to production (Section 11)
Integration with source systems	10%	5	3	All five sources are on-prem; no cross-boundary flow
Scalability and strategic fit	5%	3	5	AWS scales by configuration; on-prem is ample for T3
Weighted score	100%	4.25	3.45	

If the bank already has an approved AWS pattern for confidential finance data with on-prem sources, regulatory readiness, time-to-value and TCO each rise by one point, taking AWS to about 3.95 and making the options close to equal.

10. Risk, regulatory and control considerations

The material risks for an investment bank back office are control and dependency risks, not capacity. Each has a named mitigation.

Risk	Applies to	Impact	Mitigation
Event lost or processed twice	Both	Wrong READY / BLOCKED status used for sign-off	Persist before acknowledge, inbox table, reconciliation of source counts vs decision log
Direct Connect outage between DC and AWS	AWS	No new decisions while sources keep running	Redundant DX links, source-side retry buffer, degraded-mode runbook
Cloud data classification or approval delay	AWS	Go-live slips by months	Engage cloud risk and CISO at design stage; confirm data classes early
Regulator expectation of cloud exit strategy and concentration risk	AWS	Mandatory documented exit plan and testing	Platform-neutral design (Section 5), Terraform, PostgreSQL portable schema
Oracle to PostgreSQL migration	AWS	Dialect rework, new DBA skills, tooling change	Use JPA and portable SQL; decide engine before build starts
Shared platform capacity or change freeze	On-prem	Scaling or patching waits on other teams	Pre-book T2 capacity; agree OLA with infra teams
Hardware refresh and version end-of-life	On-prem	Unplanned migration effort in Years 3 to 5	Track platform roadmap; containerised deployment eases moves
SOX / ITGC evidence gaps	Both	Audit findings on a finance control platform	Automated release evidence, segregation of duties, access recertification
Operational resilience (important business service mapping, impact tolerance)	Both	OneFin may be classed as supporting a critical finance process	Annual DR test, documented impact tolerance, dependency mapping

Risk	Applies to	Impact	Mitigation
Audit retention and legal hold	Both	Inability to explain a past decision	7-year archive with immutable storage (WORM) and replay tooling

OneFin holds status and metadata rather than trade or client data, which should keep it at a lower data classification. This needs confirmation from the data owner, as it strongly affects AWS approval effort.

11. Delivery timeline and time-to-value

On-premises reaches production in about 10 weeks; AWS takes about 20 weeks even with an existing landing zone. Application build runs in parallel in both cases and is not the constraint.

Workstream	On-premises	AWS (landing zone exists)
Approvals: architecture review, security, data classification	Weeks 1 to 3	Weeks 1 to 8 (adds cloud risk assessment and exit plan)
Environment provisioning (DEV, SIT, UAT, PROD, DR)	Weeks 2 to 6 via standard requests	Weeks 4 to 12: account vending, IaC, VPC endpoints, DX routes, firewall rules
Messaging and database setup	Weeks 3 to 6: Kafka topics, Oracle schemas	Weeks 6 to 12: SQS, Aurora, PostgreSQL schema adaptation
CI/CD and observability	Weeks 4 to 7, existing pipelines	Weeks 8 to 14, new cloud pipelines and log forwarding
Performance, DR and penetration testing	Weeks 7 to 9	Weeks 14 to 18
Change approval and go-live	Week 10	Week 20

A 10-week gap means about 2.5 months earlier benefit from unified readiness tracking. Management should weigh that against the long-term operating model it wants.

12. Recommendation and next steps

Recommendation: deploy Phase 1 on-premises, built cloud-portable, and formally revisit AWS at the 12-month review or when a trigger below occurs.

Reasons:

1. At 300 events per minute the workload is small; neither option has a capacity advantage that matters.
2. All source systems are on-premises, so on-prem avoids a new network dependency and cross-boundary approval.
3. Production is about 10 weeks sooner, with a lower base-case 3-year TCO.
4. Portable design (Section 5) keeps a later move to AWS a platform change, not a rewrite.

Triggers to revisit AWS

- Sustained volume above T2 (25/s) or scope expansion into higher-volume workloads
- Two or more source systems move to AWS
- An approved AWS pattern exists for confidential finance data with on-prem sources
- On-prem chargeback or Oracle licensing costs rise materially above the Section 8.2 assumptions
- Enterprise cloud-first mandate for new platforms

Next steps

- ☐ Replace placeholder chargeback rates with Infrastructure Finance figures
- ☐ Confirm data classification of OneFin events with the data owner
- ☐ Confirm whether the Oracle estate is under an enterprise agreement or ULA
- ☐ Agree availability, RPO and RTO targets with the business owner and operational resilience
- ☐ Run a 1-week load test at 5/s, 25/s and 50/s on the Spring Boot proof of concept
- ☐ Present this evaluation to the architecture review board for decision

Open questions

- How many outcomes are tracked per business day, and what is the average fan-out per event?
- Is there an existing AWS landing zone approved for finance back office workloads?
- Does the enterprise Kafka platform have spare partition and storage capacity for T3?
- Is audit retention 7 years, or longer under local record-keeping rules?

Appendix: pricing assumptions and sources

All AWS prices are us-east-1 on-demand list prices as of September 2026, before enterprise discount; other regions are typically 10% to 30% higher. Validate with the AWS Pricing Calculator before budget approval.

Item	Unit price used	Source
Fargate vCPU / memory (Linux x86)	\$0.04048 per vCPU-hour, \$0.004445 per GB-hour	AWS Fargate pricing , LeanOps summary
Aurora PostgreSQL db.r6g.large	~\$0.26 per hour; db.r6g.xlarge ~\$0.52	Amazon Aurora pricing , Holori r6g.large
Aurora storage and I/O (Standard)	\$0.10 per GB-month, \$0.20 per million I/O	Pump Aurora guide
SQS FIFO requests	\$0.50 per million; batching up to 10 messages per request	CloudBurn SQS pricing
Oracle Database Enterprise Edition	\$47,500 per processor, 22% annual support	Redress Oracle licensing guide
Oracle Partitioning option	\$11,500 per processor	Reveal Compliance Oracle cost
Oracle standby licensing	Physical standby licensed same as production	Redress Active Data Guard
Blended FTE cost	\$100K per FTE-year, onshore/offshore mix	Assumption, to be confirmed by Finance
On-prem chargeback rates	\$35 per vCPU-month, \$0.60 per GB-month, DBaaS \$1,500 to \$3,000 per month	Placeholders, to be replaced by Infrastructure Finance
Cloud platform chargeback	\$500 per month	Assumption, to be confirmed by cloud platform team